Deployment

This is how Candex is built, shipped, and operated. The API runs as a **Docker Compose** stack per environment; the mobile app ships to the stores with **Fastlane**. To run it on your machine, see Local Installation; for the system around it, Architecture & Overview.

Environments

Each environment is its own Compose stack (an API container + a PostgreSQL 17 container):

Environment	Compose file	Exposure
local	`compose.local.yaml`	Throwaway stack via `scripts/candex.sh` (Local Installation)
dev	`compose.dev.yaml`	`api-dev` + `db-dev`; API on host `8081`, debugger on `5005`
prod	`compose.prod.yaml`	`api-prod` + `db-prod`; **no published ports** — reached through the reverse proxy

Each API container runs with `SPRING_PROFILES_ACTIVE` set to its environment, restarts `unless-stopped`, and keeps rotated `json-file` logs.

The API image

The API is built by a multi-stage `Dockerfile` — compiled with Maven, then run on a lean JRE:

```
FROM maven:3.9.9-eclipse-temurin-21 AS build
# ... mvn -B -DskipTests package
FROM eclipse-temurin:21-jre
COPY --from=build /app/target/*.jar /app/app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "/app/app.jar"]
```

Configuration is entirely environment-driven (database, security, SQL console, Uploadcare) — see Local Installation for the variables. Nothing secret is baked into the image.

Operating a stack

`scripts/candex.sh` operates a running dev or prod stack from the server — without stopping it:

```
./scripts/candex.sh psql <dev|prod>        # a psql session
./scripts/candex.sh logs <dev|prod>        # follow the API logs
./scripts/candex.sh dump <dev|prod>        # pg_dump the database
./scripts/candex.sh exec <dev|prod> "<SQL>"
```

CI/CD

GitHub Actions drive integration and delivery (`.github/workflows`):

Workflow	Role
`ci.yml`	Quality gate on push / PR — build, tests, lint (and Qodana)
`deploy-dev.yml` • `deploy-prod.yml`	Deploy the API to dev / prod
`backup-prod.yml`	Scheduled production database backup

The dashboard has its own `ci.yml` quality gate.

Reverse proxy & TLS

> **To confirm**
>
> The production API publishes no ports; the public entrypoint and TLS are handled by a **reverse proxy at the server level** (a `CADDY_EMAIL` variable points to Caddy). The exact proxy configuration lives outside these repositories and is not documented here yet.

Mobile app releases

The app ships to the **Play Store** and **App Store** with **Fastlane**, triggered manually from GitHub Actions. Version bump and changelog stay local and human-reviewed — CI only deploys an existing tag:

```
npm run release 1.2.0            # bump + annotated tag x.y.z (message = changelog)
git push && git push origin 1.2.0
# GitHub ▸ Actions ▸ "Deploy to Stores" ▸ Run workflow (store: both / play / appstore)
```

`deploy.yml` (a manual `workflow_dispatch`) builds the chosen tag and submits it through the Fastlane lanes `android deploy` (AAB → Play production) and `ios deploy` (IPA → App Store Connect).

Where to go next

- Database backups → **Backups** *(planned)*
- Run it on your machine → Local Installation
- The API itself → Back-end